

ACUTE TRANSCRIPTIONAL RESPONSE · BULK RNA-SEQ TIME COURSE

A Rapid, Coordinated and Transient Transcriptional Program Triggered by Acute Netrin-1 Stimulation in Cortical Neurons

5- and 15-minute Netrin-1 time course in E18 rat cortical neurons

Thesis Defense

[Your name] · [Department / Institution] · [Date]

Outline

- ▶ Background — Netrin-1 and the gap in acute transcription
 - Why look at minutes, not hours?
- ▶ Question, hypothesis & experimental design
- ▶ Analysis pipeline & quality control
- ▶ Results
 - 1 · The 5-minute response (genes)
 - 2 · Transient dynamics (IEGs, ROCK1)
 - 3 · Pathway-level coordination (GSEA)
 - 4 · Return to baseline — the $r = -0.99$ mirror
- ▶ Discussion, limitations & conclusions

THE ONE-SENTENCE STORY

A 5-min Netrin-1 pulse activates an axon-guidance / projection program and suppresses RNA-processing & ribosome biogenesis — then by 15 min returns to baseline with $r = -0.99$ precision.

BACKGROUND

Netrin-1: a classic guidance cue — acting in seconds

► **Netrin-1 is a prototypical axon-guidance molecule.**

- Steers the growth cone via DCC and UNC5 receptors.

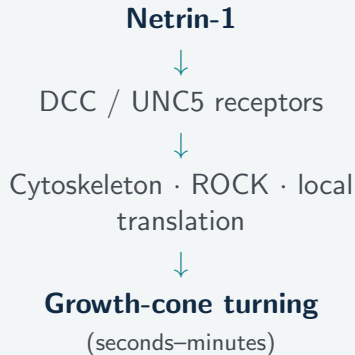
► **Canonical mechanism is POST-translational and LOCAL.**

- Cytoskeletal remodeling, local translation.
- Seconds-to-minutes timescale.

► **The field focuses on protein-level signaling. . .**

- . . . and overlooks the nucleus on this timescale.

CANONICAL NETRIN-1 SIGNALING



BACKGROUND

The knowledge gap: minutes-scale transcription



- ▶ Almost all Netrin transcriptome studies sample **hours to days** after stimulation.
- ▶ **The acute, minute-scale transcriptional response is essentially uncharted.**
- ▶ Open questions: does the transcriptome change within minutes? *How?* And is the change *transient* or *sustained*?

AIM

Research question & hypothesis

RESEARCH QUESTION

Does acute Netrin-1 stimulation rapidly reshape the neuronal transcriptome — and is that response transient or persistent?

HYPOTHESIS

Netrin-1 induces a fast, coordinated transcriptional response ...that is **transient** — switched on, then off — rather than a stable reprogramming.

APPROACH

Bulk RNA-seq time course 0/5/15 min · DE-Seq2 differential expression · GSEA pathway coordination · compare 5v0 vs 15v5 to test reversibility.

METHODS

Experimental design

- ▶ **E18 rat cortical mixed primary cultures.**
- ▶ **Three conditions:**
 - Vehicle (PBS, 5 min) — reference / 0 min
 - Netrin-1, 5 min · Netrin-1, 15 min
- ▶ **3 biological replicates → 9 samples.**
- ▶ **Design limitation:** one shared PBS-5min control serves *both* Netrin timepoints — no independent PBS-15min.
 - confounds 15v0; main results use 5v0 and 15v5.

9 SAMPLES = 3 COND. × 3 REP.

Vehicle / 0

rep 1

rep 2

rep 3

Netrin 5 min

rep 1

rep 2

rep 3

Netrin 15 min

rep 1

rep 2

rep 3

METHODS

Analysis pipeline



Quantification with DRAGEN Salmon (gene level); counts imported via tximport with a length offset into DESeq2. Two-pass filtering keeps **14,161 of 32,623** genes (85.6% SYMBOL-mapped). Replicate is a batch term; **batch correction is for visualization only — DE tests use raw counts.**

METHODS

Statistical model & significance

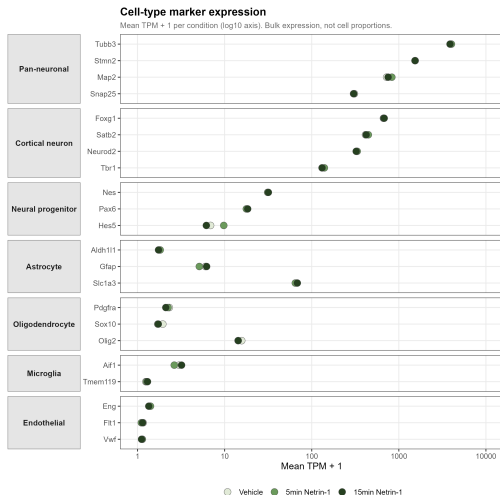
DIFFERENTIAL EXPRESSION

- ▶ Model: $\sim \text{replicate} + \text{timepoint}$ (ref = Vehicle).
- ▶ Replicate dominates variance: PCA **rep** $\approx 52\%$, **time** $\approx 34\%$.
- ▶ Wald: 5v0, 15v0, 15v5; LRT (reduced $\sim \text{rep}$).
- ▶ **RELAXED**: $\text{padj} < 0.05$.
- ▶ **STRICT**: $\text{padj} < 0.05$ & $|\log_2\text{FC}| > 0.585$ ($\text{FC} > 1.5$).

GENE-SET ENRICHMENT (GSEA)

- ▶ `clusterProfiler gseGO(BP) + gseKEGG(rno)`.
- ▶ Rank = DESeq2 Wald stat; entrez de-dup ($\max |\text{stat}|$).
- ▶ $\text{minGSSize } 15 \cdot \text{maxGSSize } 500 \cdot \text{BH} \cdot \text{seed } 123$.
- ▶ GO simplified (0.7) to drop redundancy.
- ▶ **Why GSEA: acute signal weak per-gene — pathway aggregation reveals the coordinated program.**

Quality control I — neuron-dominated & stable



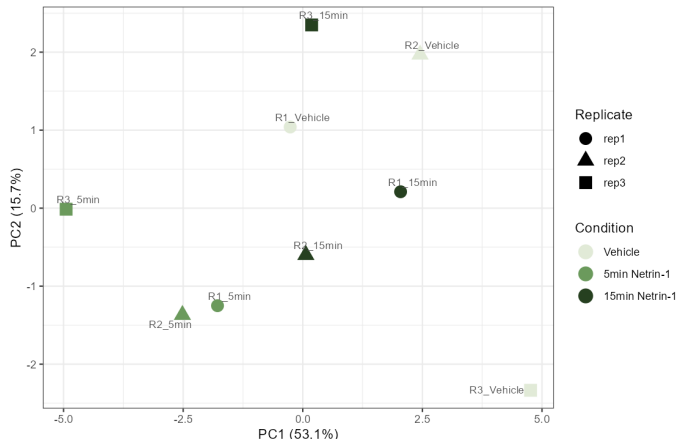
- ▶ **Pan-neuronal markers dominate:**
Tubb3 ≈ 3900 , Stmn2 ≈ 1540 , Map2 ≈ 750 TPM.
– 2–4 orders above glial/endothelial (Vwf, Flt1 ≈ 0.1).
- ▶ **Marker profile near-identical across all 3 conditions.**
→ cell composition does *not* change.
- ▶ **Downstream changes = genuine transcriptional response, not a shift in cell types.**

RESULTS · QC

Quality control II — replicate is the top batch effect

PCA (top-500 var genes, VST, AFTER batch correction)

removeBatchEffect(batch=replicate, design=~timepoint) — visualization only; DE testing uses raw counts
PC1 53.1% | PC2 15.7% | PC3 8.7% | PC4 8.5%



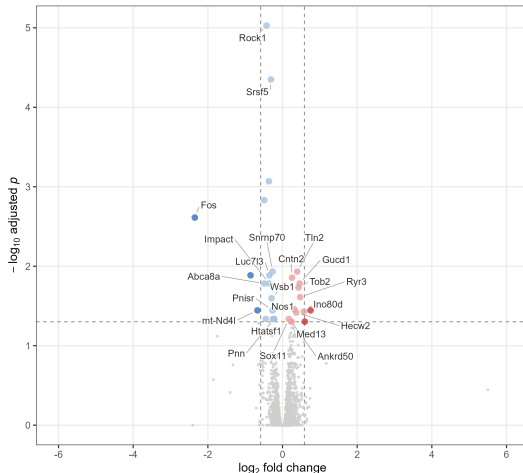
- ▶ **Top variation = biological REPLICATE, not treatment** (rep $\approx 52\%$, time $\approx 34\%$).
- ▶ Handled: replicate a batch term; limma removeBatchEffect for visualization only.
- ▶ **After correction, PC1 mainly reflects TIMEPOINT** — samples separate by 0/5/15 min.

RESULTS · DEGS

Result 1 — the 5-minute response is real but small

Netrin-1 (5 min) vs Vehicle

Netrin-1 (5 min) vs Vehicle · 32 significant (padj<0.05); 5 Strict (2 up + 3 down); 23 labeled (symbol only)



► **5v0: RELAXED 32** (14↑/18↓);
STRICT 5 (2↑/3↓).

► **Fos stands out:** log₂FC -2.35,
padj 2.4e-3 — strongly down.

► Most effects modest — a fast,
subtle, pathway-level signal.

→ **motivates the GSEA analysis.**

Core 5-minute DEGs map onto coherent modules

Gene	log2FC	padj	Module
Fos	−2.35	2.4e−3	Immediate-early gene (IEG ↓)
Rock1	−0.43	9.4e−6	Cytoskeleton / ROCK
Srsf5	−0.31	4.5e−5	Splicing
Snrnp70	−0.27	1.2e−2	Splicing
Cntn2	+0.25	1.4e−2	Axon guidance
Nos1	+0.32	3.5e−2	NO–calcium
Ryr3	+0.47	2.4e−2	NO–calcium
Tln2	+0.39	1.2e−2	Adhesion

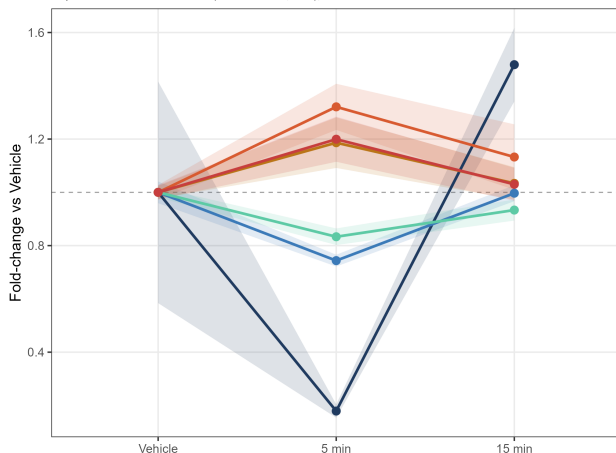
Down: IEG, cytoskeleton, splicing. Up: axon guidance, NO–calcium, adhesion. Small individually
— but clean, interpretable modules.

RESULTS · DYNAMICS

Result 2 — a transient 5-minute pulse

Acute time-course trajectories (Netrin-1)

Expression relative to Vehicle (mean $\pm$ SEM, n=3); dashed line = Vehicle baseline



► **LRT: 19 genes change significantly over time.**

► **Dominant shape = a 5-min PULSE:** depart at 5 min, return by 15.

► **First direct evidence the program is transient, not sustained.**

► **Caveat:** Fos at 15 min overshoots baseline; variable Vehicle replicates → wide error band.

RESULTS · DYNAMICS

A surprise: two immediate-early genes are SUPPRESSED

Immediate-early genes are normally activated by neuronal activity. Here, acute Netrin-1 does the opposite — it acutely DOWN-regulates them.

Fos

$\log_2FC -2.35 \cdot \text{padj } 2.4e-3$

The strongest DEG of the study.

Strong, significant acute repression.

Npas4

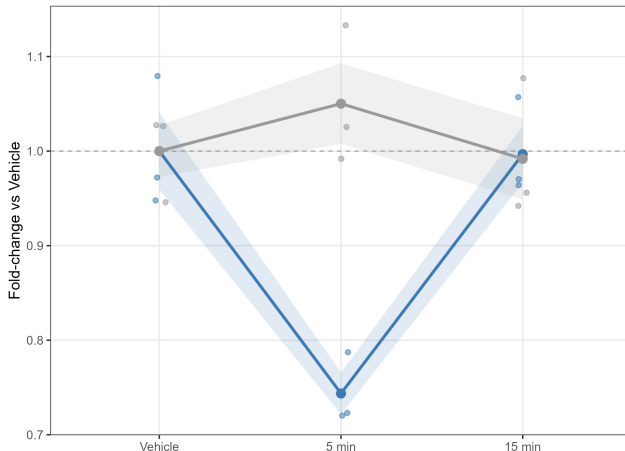
$\log_2FC -1.75 \cdot \text{padj } 0.076$ (Wald, n.s.)

Significant by the time-course LRT.

A second acutely-repressed IEG.

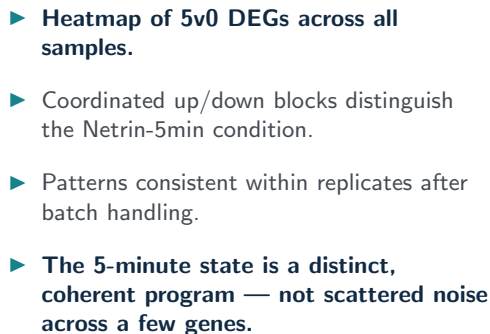
Selective control: Rock1 down, Rock2 untouched

ROCK1 vs ROCK2: acute time-course (Netrin-1)

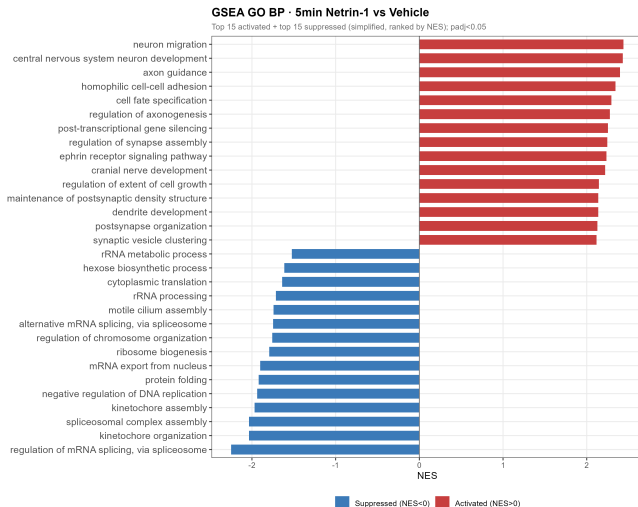
Fold-change vs Vehicle (mean $\pm$ SEM, n=3). 5 min vs Vehicle padj: Rock1 9.4e-06 (sig) | Rock2 1.00 (n.s.)

- ▶ **Rock1:** log2FC -0.43 , padj $9.4e-6$ — down.
- ▶ **Rock2:** log2FC $+0.07$, padj ≈ 1.0 — unchanged.
- ▶ **Two paralogues, opposite fates in one cell.**
- ▶ ROCK1 is a classic Netrin–DCC effector.
- **isoform-selective transcriptional control — precise, not blunt.**

DEGs separate the conditions coherently



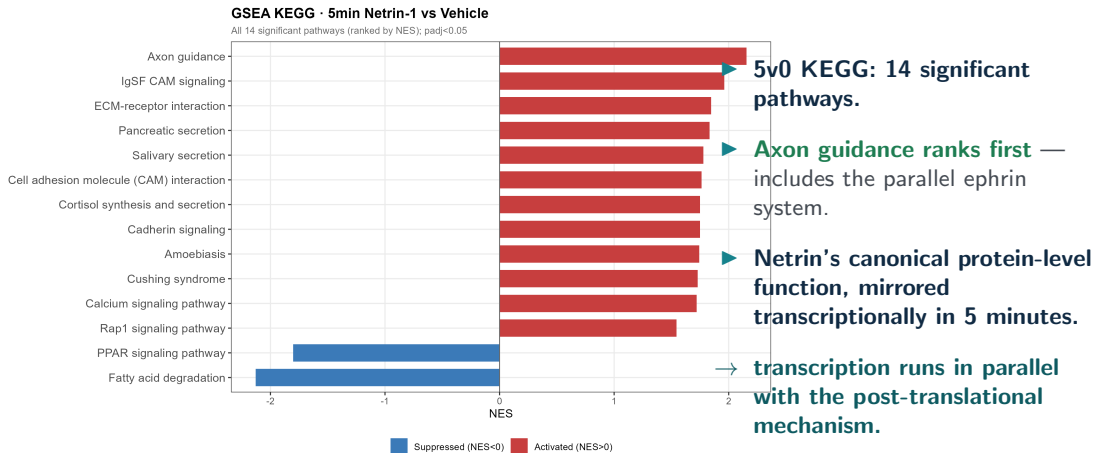
Result 3 — GSEA reveals a coordinated program



- **5v0 GO-BP: 313 significant (285↑/28↓).**
- **ACTIVATED:** axon guidance, neuron projection, synapse assembly.
- **SUPPRESSED:** mRNA splicing/processing, ribosome biogenesis & translation.
- **GSEA elevates single-gene modules into a genome-wide coordinated signal.**

RESULTS · PATHWAYS

KEGG: axon guidance tops the activated pathways

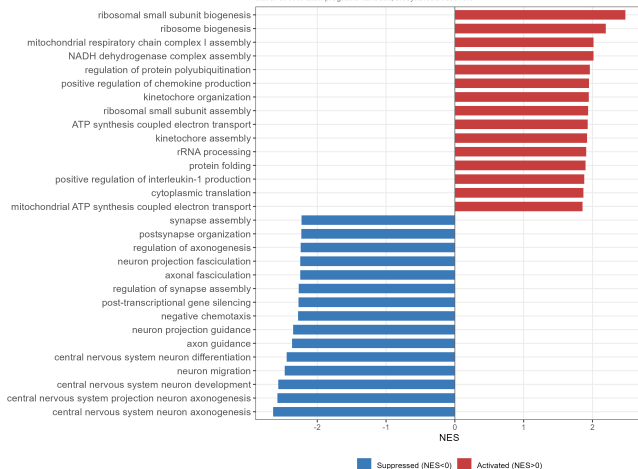


RESULTS · RETURN

Result 4 — by 15 min the program fully reverses

GSEA GO BP · 15min vs 5min Netrin-1 (time evolution)

Top 15 activated + top 15 suppressed (ranked by NES); padj<0.05
Mirror of 5v0: axon programs fall back, biosynthesis recovers



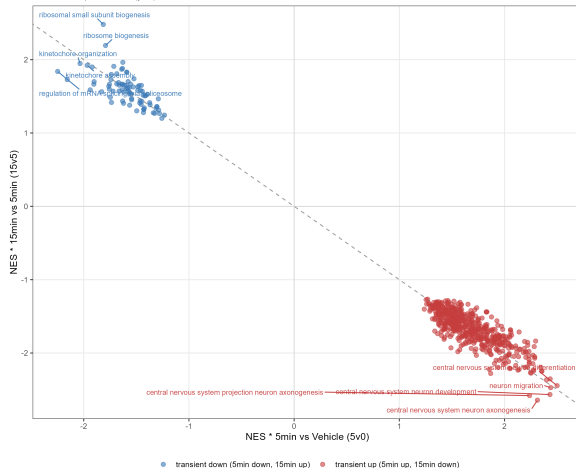
- ▶ **15v5 GO-BP: 295 significant (31↑/264↓).**
- ▶ **Direction FLIPPED vs 5v0:** guidance ↓, biosynthesis ↑ (recover).
- ▶ **15v0 GO-BP: 0 significant** → back at baseline by 15 min.
- ▶ **Switched on at 5 min, off by 15.**

RESULTS · RETURN

The signature result: an $r = -0.99$ mirror

GSEA pathway mirror: 5v0 vs 15v5 (GO BP)

Each point = one pathway. NES correlation $r = -0.99$ (negative = transient then reversal)
Dashed line = perfect mirror ($y = -x$)



$$r = -0.99$$

5v0 vs 15v5 pathway NES correlation

- ▶ 603 pathways shared between 5v0 and 15v5.
- ▶ Their enrichment scores are near-perfect opposites.
- ▶ Every pathway up at 5 min is suppressed 5→15 min by almost the same amount.

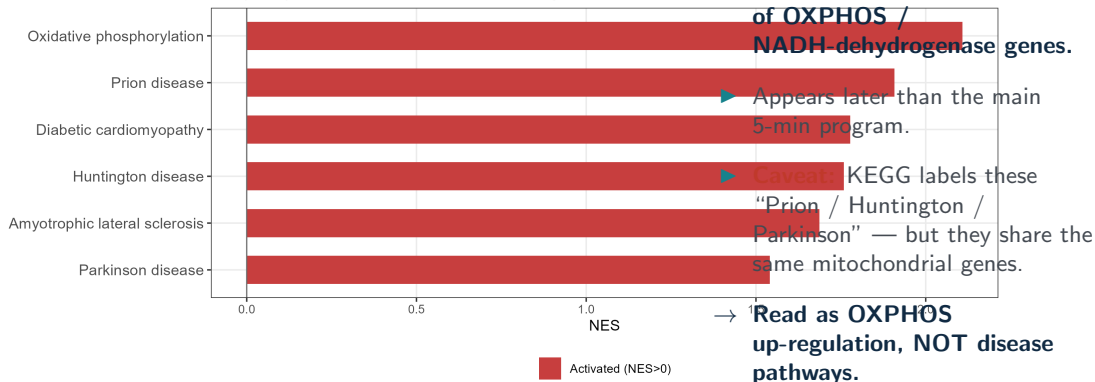
→ quantitative proof of a precisely-undone transient.

RESULTS · RETURN

A late, secondary signal: mitochondrial OXPHOS

GSEA KEGG · 15min Netrin-1 vs Vehicle

All 6 significant pathways; padj<0.05 (GO BP: 0 significant — signal returned to baseline)
Note: pathway names reflect shared mitochondrial/OXPHOS gene sets



SYNTHESIS

Modules of the transient program

Module	5 min	→ 15 min	Evidence
Splicing / RNA processing	down	recover	DEG + GSEA + LRT
Ribosome / translation	down	recover	GSEA + mirror
Immediate-early genes	down	rebound	Fos, Npas4 (LRT)
Cytoskeleton / ROCK	down	recover	Rock1 (Rock2 n.s.)
Axon guidance / projection	up	fall back	DEG + GSEA + KEGG
Synapse assembly	up	fall back	GSEA
NO-calcium signaling	up	fall back	Nos1, Ryr3 + KEGG
Adhesion	up	fall back	Tln2 + KEGG
Mitochondria / OXPHOS	flat	late rise	GSEA 15v5 / 15v0

Guidance/projection UP, biosynthesis & IEGs DOWN at 5 min — all reversing by 15 min, with a late OXPHOS rise.

SYNTHESIS

A model: switch on at 5 min, off by 15

UP: axon guidance · projection · synapse · NO–calcium · adhesion

DOWN: splicing · ribosome biogenesis · translation · IEGs · ROCK1



A 5-minute Netrin-1 pulse drives a coordinated transcriptional swing, then near-perfectly reverses it ($r = -0.99$) by 15 minutes — a transient program, with a late OXPHOS rise as the one lingering signal.

DISCUSSION

Discussion — five take-home messages

- 1 Dual IEG suppression** — Fos AND Npas4 acutely down, opposite to canonical IEG behavior. Novel.
- 2 Coordinated biosynthesis shutdown** — splicing + ribosome biogenesis + translation transiently suppressed; possible resource reallocation to local translation.
- 3 Guidance — transcriptional arm** — axon guidance induced at mRNA level in minutes, parallel to the classic post-translational mechanism.
- 4 $r = -0.99$ transient reversal** — the 5-min program precisely undone by 15 min. Strongest quantitative result.
- 5 ROCK1-selective control** — Rock1 down, Rock2 unchanged; isoform-specific regulation of a canonical Netrin–DCC effector.

DISCUSSION

Limitations — read honestly

- ▶ **n = 3, acute signal weak.** Most single-gene effects small → pathway-level GSEA is the primary analysis.
- ▶ **No independent PBS-15min control.** 15v0 confounds treatment with elapsed time; main results rely on 5v0 and 15v5.
- ▶ **Only three timepoints.** Dynamics via LRT + trend classification; dense time-series methods (e.g. ImpulseDE2) not applicable.
- ▶ **Fos shows high Vehicle-replicate variability.** Trajectory error band wide — interpreted cautiously.
- ▶ **Bulk RNA-seq, not single-cell.** Markers reflect overall expression, not cell-type proportions.

Conclusions

Acute Netrin-1 triggers a rapid, coordinated and TRANSIENT transcriptional program in cortical neurons — activating axon-guidance / projection while suppressing RNA processing and ribosome biogenesis — reversed to baseline with $r = -0.99$ precision within 15 minutes.

- ▶ First characterization of Netrin-1's minute-scale transcriptional response.
- ▶ Transcription runs in PARALLEL with the classic post-translational mechanism.
- ▶ Novel: dual-IEG suppression and ROCK1-selective regulation.
- ▶ The $r = -0.99$ mirror defines a precisely-controlled transient — not reprogramming.

Future directions

- ▶ **Denser time course** (+ proper per-timepoint controls) to map the full pulse shape.
- ▶ **Single-cell / spatial RNA-seq** to localize the response to subtypes & compartments.
- ▶ **Mechanism of dual-IEG suppression** — what represses Fos and Npas4 acutely?
- ▶ **Link the transcriptional swing to local translation** at the growth cone.
- ▶ **Test DCC/UNC5 dependence and ROCK1-selectivity** with perturbations.

Thank you

Questions & discussion welcome

One sentence to remember:

A 5-minute Netrin-1 pulse switches a coordinated transcriptional program ON — then OFF again by 15 minutes, with $r = -0.99$ precision.